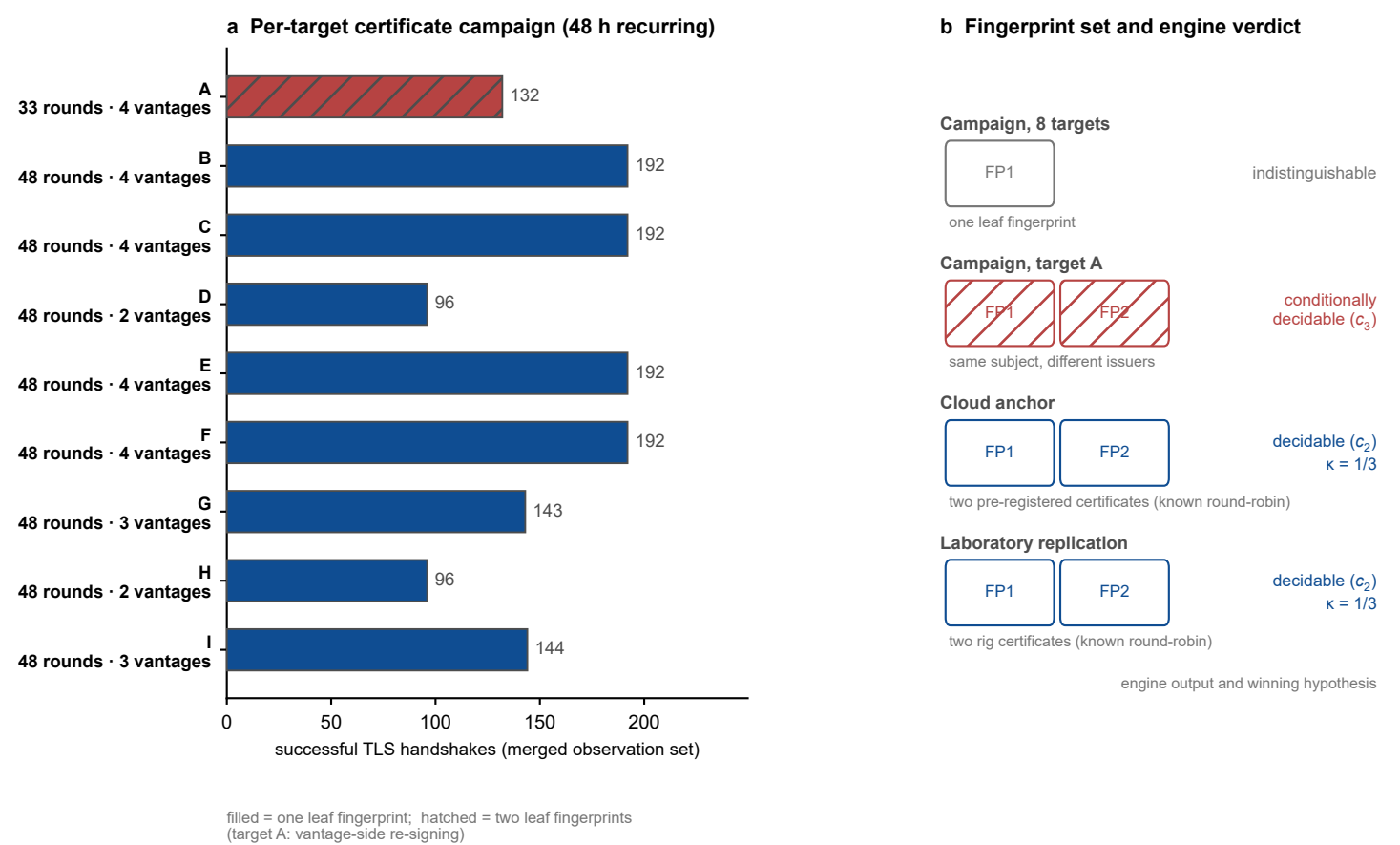

a Per-target certificate campaign (48 h recurring)
A
33 rounds · 4 vantages
132
B
48 rounds · 4 vantages
192
C
48 rounds · 4 vantages
192
D
48 rounds · 2 vantages
96
E
48 rounds · 4 vantages
192
F
48 rounds · 4 vantages
192
G
48 rounds · 3 vantages
143
H
48 rounds · 2 vantages
96
I
48 rounds · 3 vantages
144
0
50
100
150
200
successful TLS handshakes (merged observation set)
filled = one leaf fingerprint; hatched = two leaf fingerprints
(target A: vantage-side re-signing)
b Fingerprint set and engine verdict
Campaign, 8 targets
FP1
one leaf fingerprint
indistinguishable
Campaign, target A
FP1
FP2
same subject, different issuers
conditionally decidable (c_3)
Cloud anchor
FP1
FP2
two pre-registered certificates (known round-robin)
decidable (c_2)
κ = 1/3
Laboratory replication
FP1
FP2
two rig certificates (known round-robin)
decidable (c_2)
κ = 1/3
engine output and winning hypothesis